# Seungmin Lee

Department of Mathematics, Florida State University
Applied and Computational Mathematics | sl22b@fsu.edu

## Education

**Ph.D., Applied and Computational Mathematics** August 2022 - present
Florida State University, Tallahassee, Florida, USA

**M.S., Mathematics** March 2020 - February 2022
Dongguk University, Seoul, South Korea
*Thesis:* Neural Network in Solving Ordinary Differential Equations

**B.S., Mathematics** March 2016 - February 2020
Dongguk University, Seoul, South Korea

## Research Experience

**Research Assistant**, Florida State University Fall 2026; Spring 2025

**Intern**, Lawrence Livermore National Laboratory (LLNL)
June 3 - July 13, 2024

## Publications and Manuscripts

### Published

[1] **Seungmin Lee** and Sanghyun Lee.
Convergence and numerical simulations of the continuous data assimilation for Biot's poroelasticity system.
*Computers & Mathematics with Applications*, 202:20-37, 2026.
doi:10.1016/j.camwa.2025.11.008

### Submitted

[2] Sanghyun Lee, **Seungmin Lee**, and Chen Liu.
An optimization-based property-preserving limiter for two-phase flow in porous media on non-matching meshes.
*Submitted.* arXiv:2610.33940

[3] Chen Liu, **Seungmin Lee**, and Sanghyun Lee.
Optimization-based cell average limiters as metric projections: Constrained evolution and convergence analysis for convection-diffusion equations.
*Submitted.*

### In Preparation

[4] Feng Bao, Phuoc Toan Huynh, Sanghyun Lee, and **Seungmin Lee**.
Ensemble score filtering for state estimation in a two-phase flow model with rare earth materials.
*Work in progress.*

## Teaching Experience

### Florida State University

### Course Instructor

| | |
|---|---|
| Calculus II | Spring 2026 |
| Calculus I | Fall 2025 |

### Teaching Assistant

| Course | Role | Term(s) |
|---|---|---|
| Discrete Mathematics II | Recitation TA | Fall 2024 |
| Calculus I | Recitation TA | Spring 2024 |
| College Algebra | Lab TA | Fall 2022, Spring 2023, Fall 2023 |
| Liberal Arts Mathematics | Lab TA | Fall 2023 |
| Analytic Trigonometry | Lecture TA | Summer 2023 |
| Business Calculus | Lab TA | Fall 2022, Spring 2023 |

### Dongguk University

### Grading Assistant

| | |
|---|---|
| Modern Algebra | March - June 2021 |
| Cryptography | September - December 2020 |
| Calculus I and II | March 2018 - December 2019 |

## Conferences, Workshops, and Seminars

- **Applied and Computational Math Seminar**, Fall 2026.
  Florida State University, Tallahassee, Florida, USA. September 8, 2026.
- **2026 Southeast Applied and Computational Math Student Workshop**.
  Emory University, Atlanta, Georgia, USA. April 3-4, 2026.
- **M3L Team All-Hands Meeting**.
  The University of Texas at El Paso, El Paso, Texas, USA. May 15-17, 2025.
- **2025 Southeast Applied and Computational Math Student Workshop**.
  Auburn University, Auburn, Alabama, USA. April 4-6, 2025.
- **11th deal.II Users and Developers Workshop**.
  Colorado State University, Fort Collins, Colorado, USA. August 12-16, 2024.